Blockchain and Cryptocurrency

Assignment: Supply Chain Management DApp using Polygon and Hardhat

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Network: Polygon Amoy Testnet

Deployment Address: `0x6A1b64a133928054eBd66fa86820Db562227B119`

`Transaction Hash:`

`0x6e21dc4fba39583efa8b873d8ff34f1fa1490a847562633ed1524e29b93d014b`

1. System Overview

The objective of this DApp is to ensure transparency and traceability in product movement from a Manufacturer to a Consumer. The system utilizes a smart contract to record every ownership transfer on the Polygon blockchain, providing a tamper-proof audit trail.

2. Compilation and Deployment

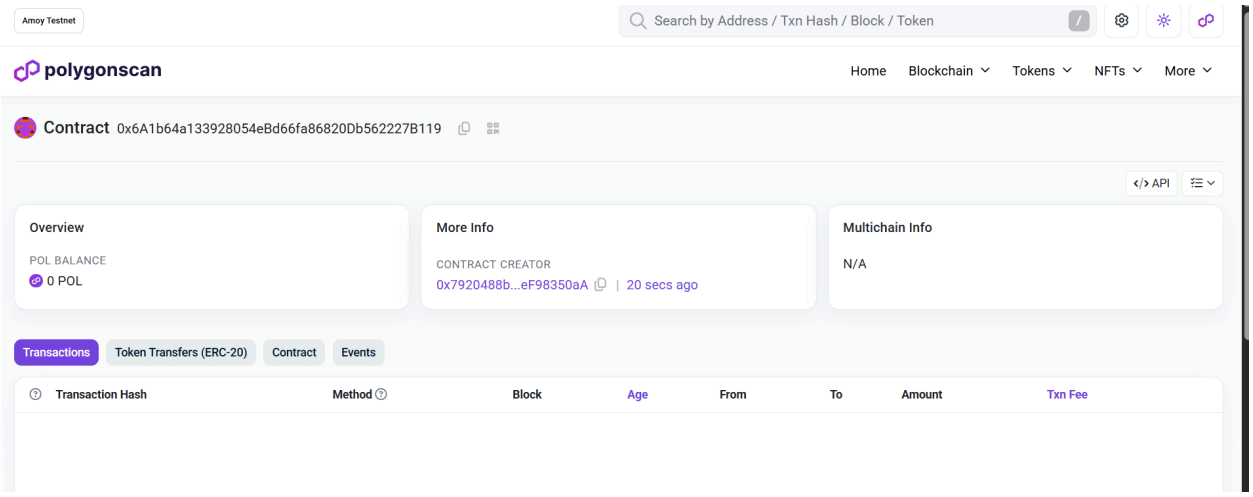
The smart contract was compiled and deployed using the Hardhat framework.

- **Compilation:** Solidity source code was converted into bytecode and ABI for frontend integration.

- **Deployment:** The contract was deployed to the **Polygon Amoy Testnet**. The deployment transaction was executed by the manufacturer's wallet (Account 2).**3. System Working and Snapshot Analysis**

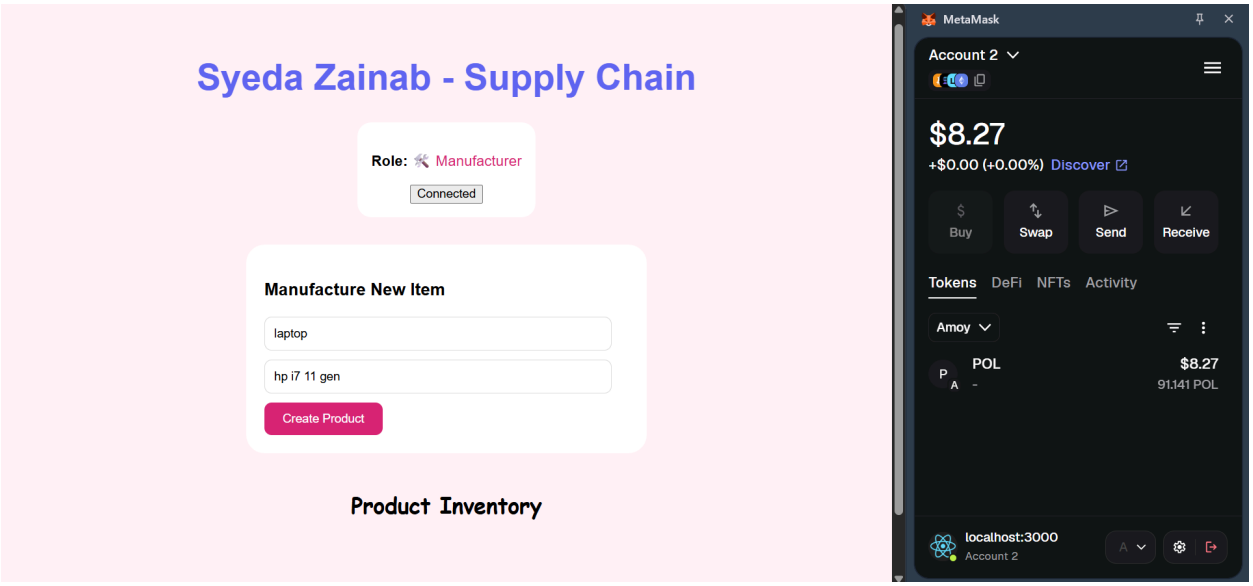
A. Contract Verification (Polygonscan)

- This snapshot(below) confirms the contract is live on the blockchain. The "Contract Creator" address is verified as 0x7920...350aA (Account 2), which acts as the system's Manufacturer



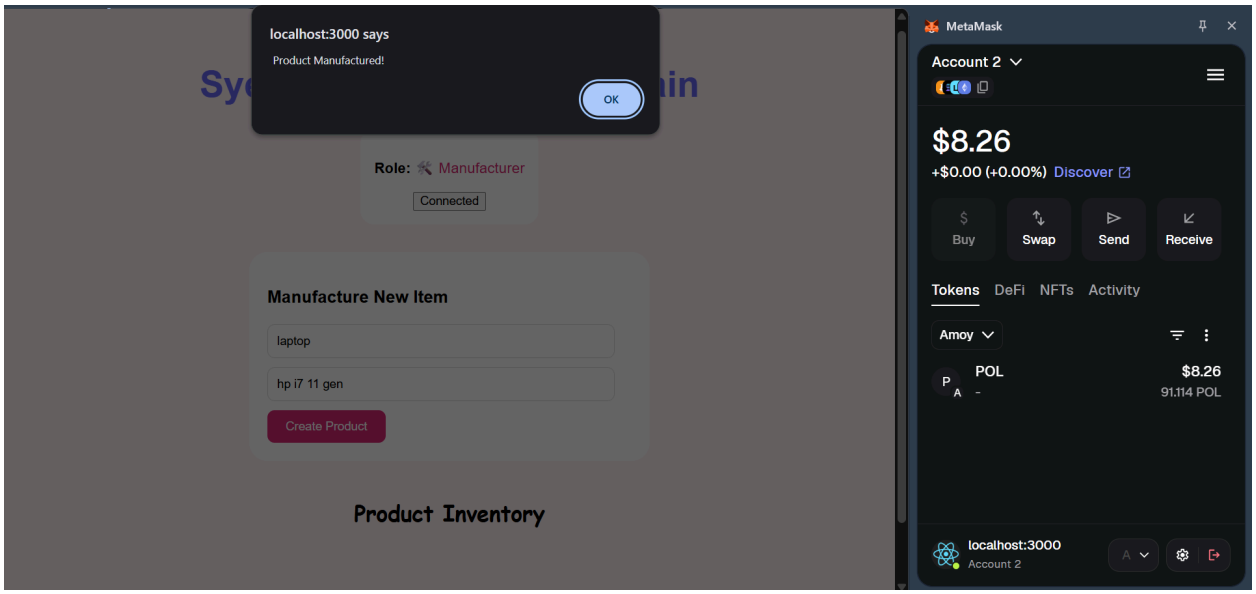
B. Product Manufacturing (Manufacturer Role)

- I logged in as the Manufacturer, initializes the process. A new product, "laptop," is being registered with a description of "hp 17 11 gen".
- MetaMask State: Account 2 holds a balance of approximately 91.141 POL to cover transaction gas fees.

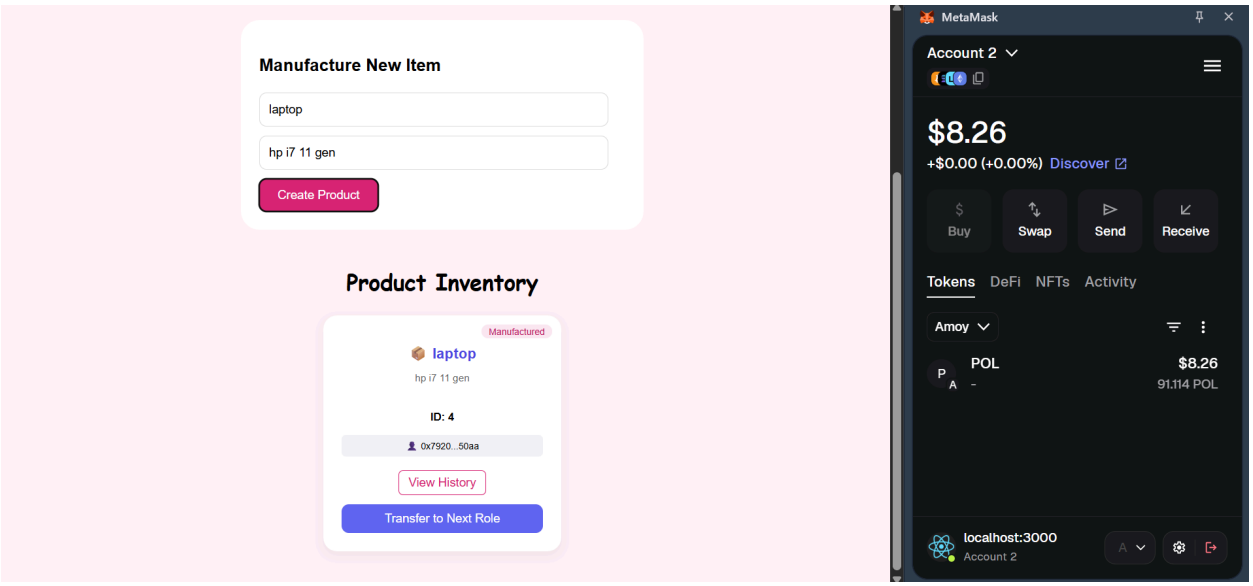


C. Registration Confirmation

- Upon confirming the transaction in MetaMask, the system provides a "Product Manufactured" alert.

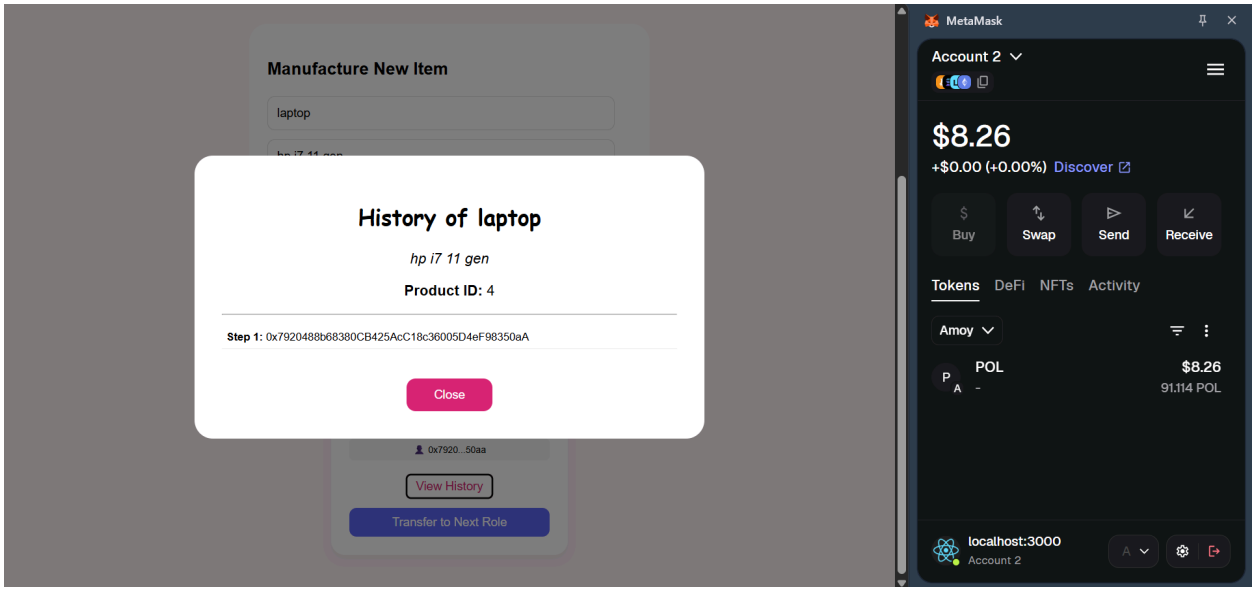


- The product is assigned a unique **ID: 4**.

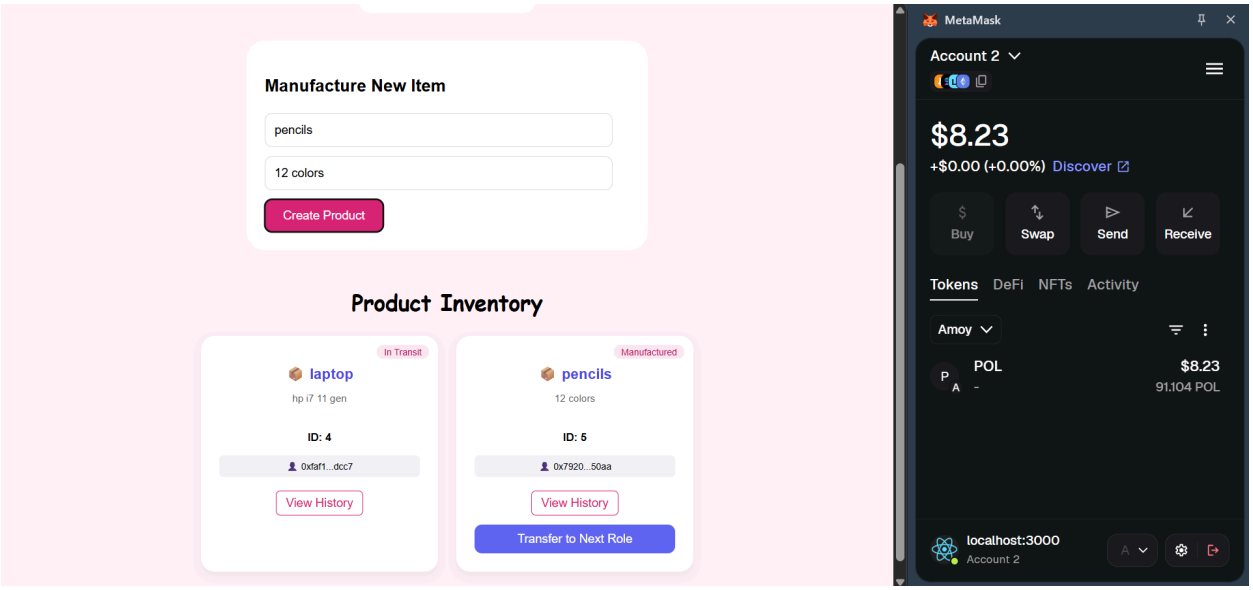


D. Audit Trail - Step 1

- Clicking "View History" shows the initial state of the product.

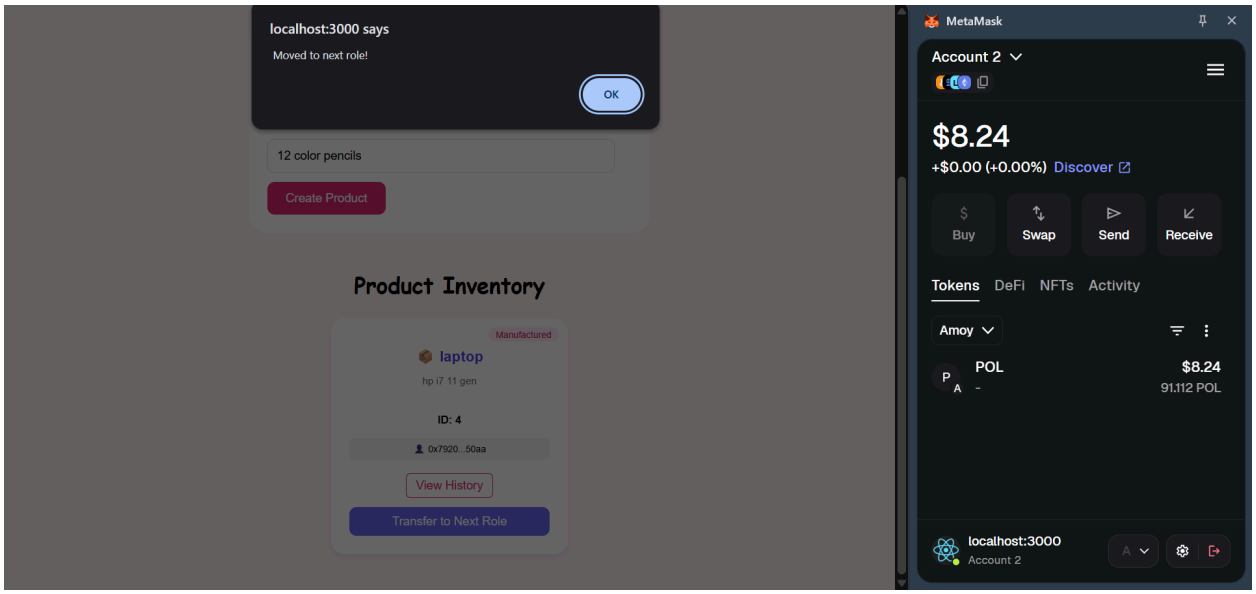


- **Step 1** records the Manufacturer's address as the original owner at the time of creation.Moreover , manufacturer added another product Pencils.

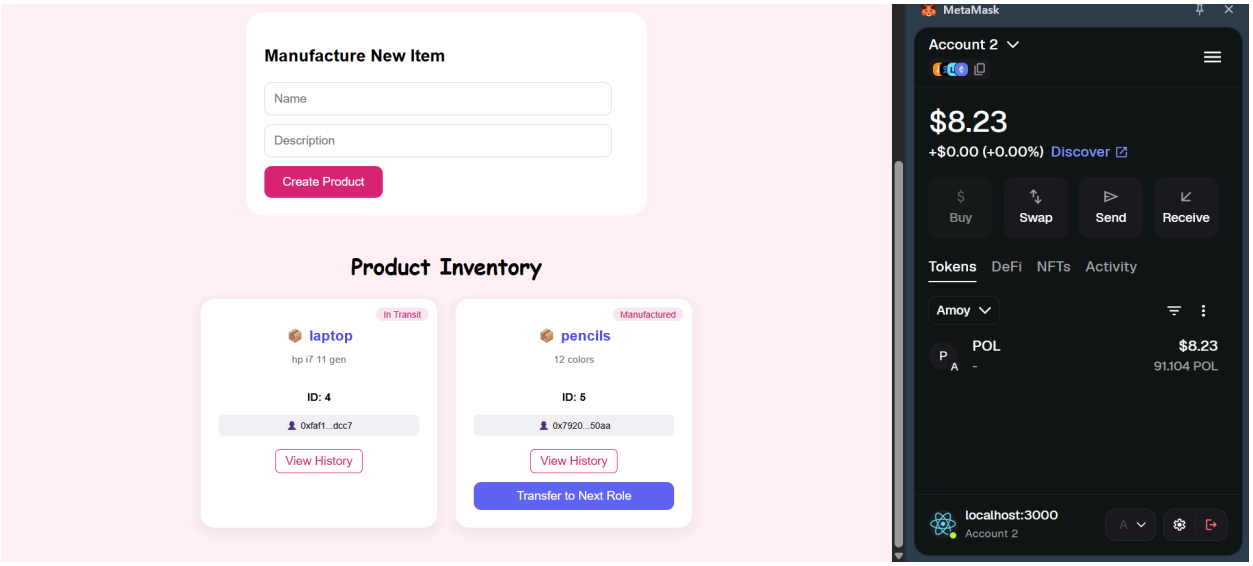


E. Transfer to Distributor

- The Manufacturer transfers ownership to the next role.

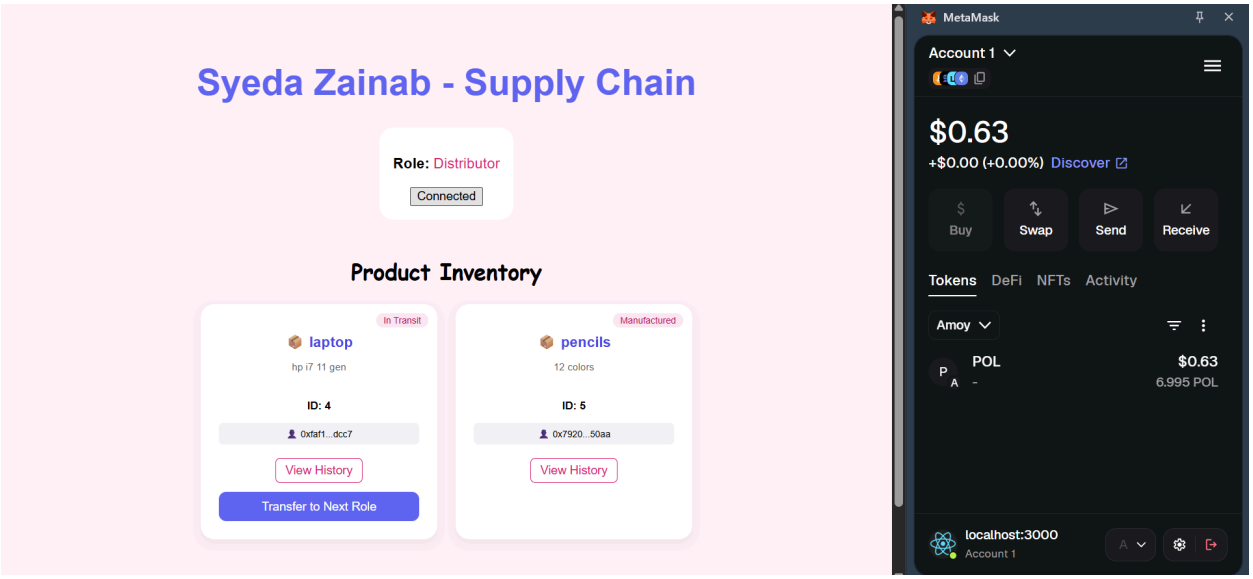


- The status of the product updates from "Manufactured" to **"In Transit"**.

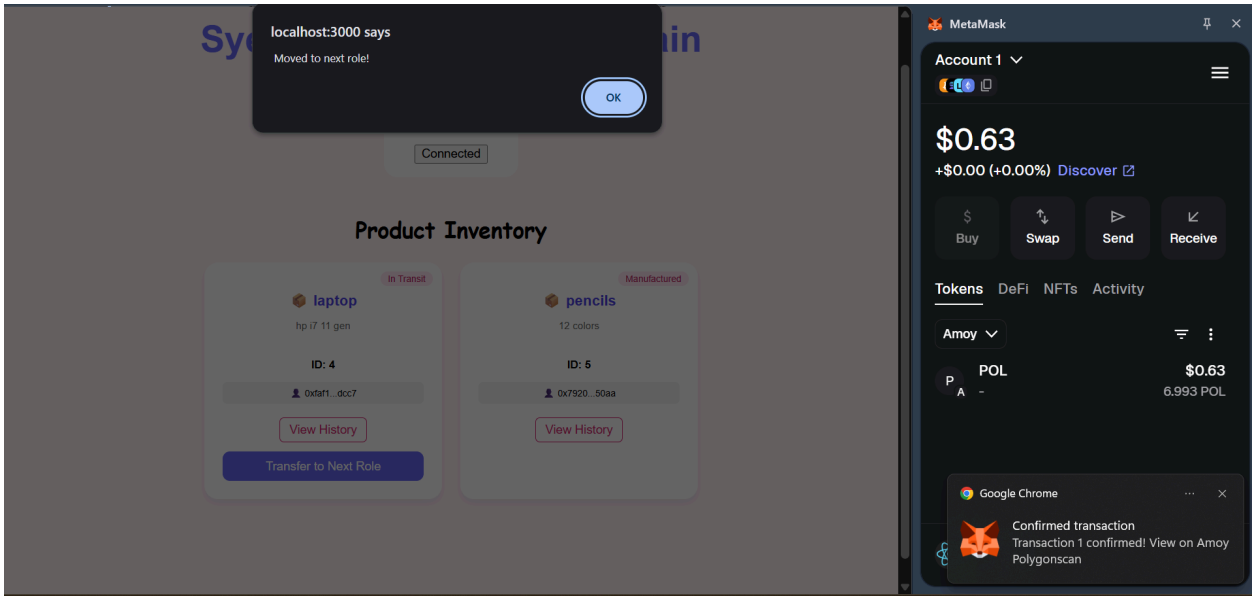


F. Distributor & Retailer Interaction

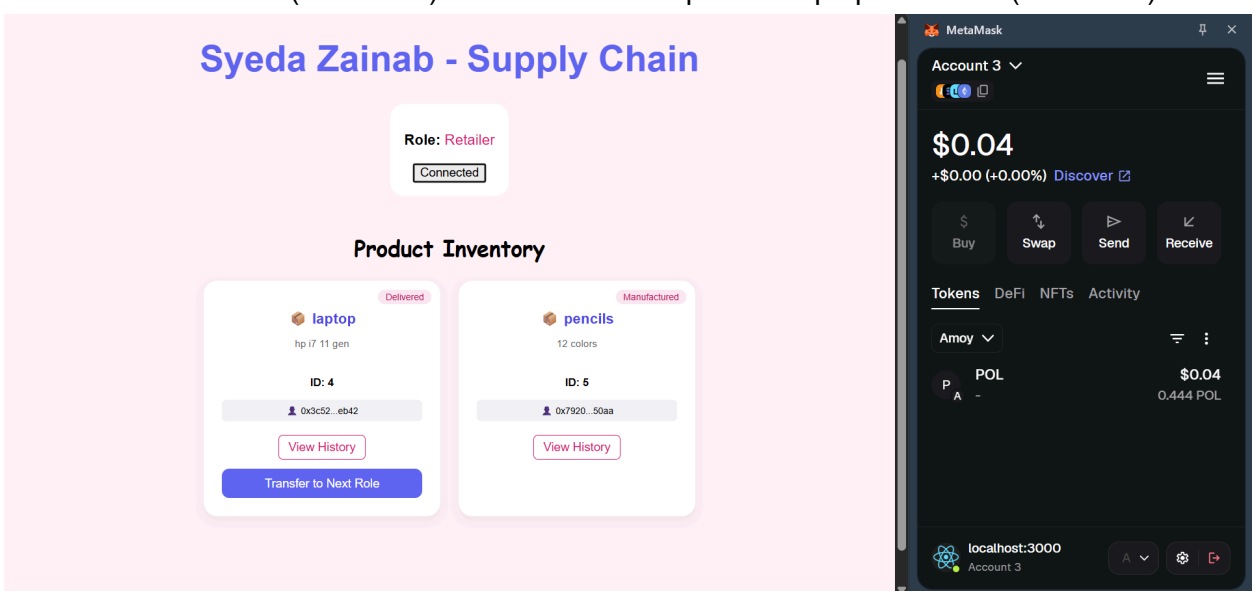
- As different users connect (Account 1 as **Distributor** and Account 3 as **Retailer**), the GUI dynamically updates the role label. This ensures that only the current owner can trigger the "Transfer to Next Role" function.



- Account 1 here acts as Distributor and it has only access to transfer laptop , it can not transfer the ownership of pencils to the next role.

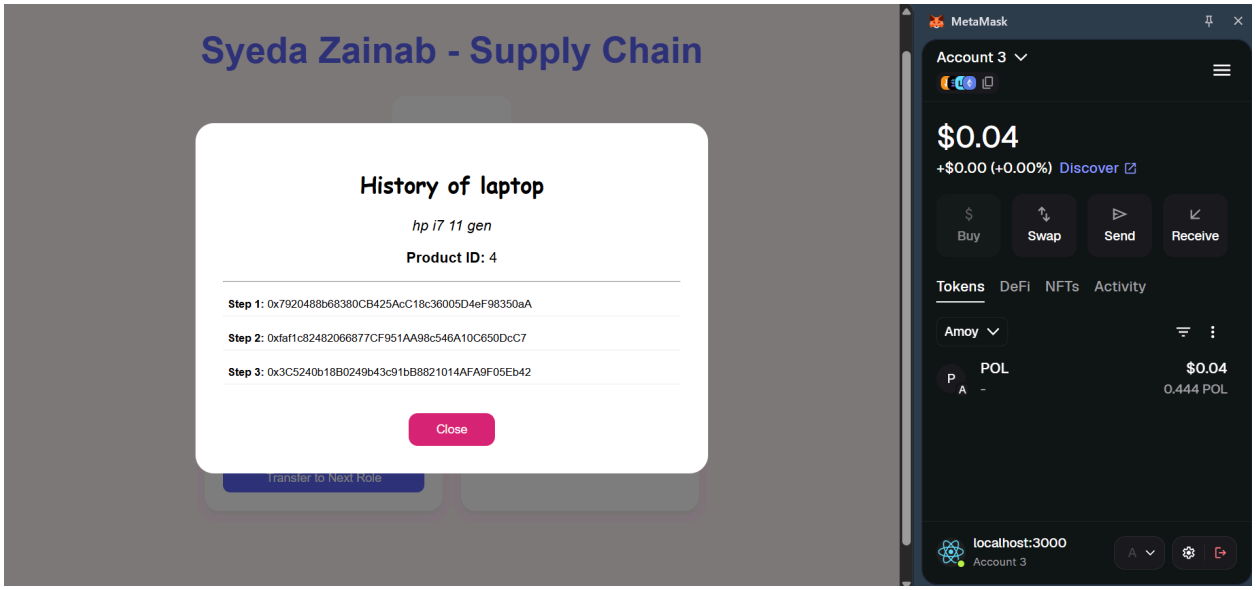


- Now account 1(distributor) transfers ownership of the laptop to retailer (account 3).

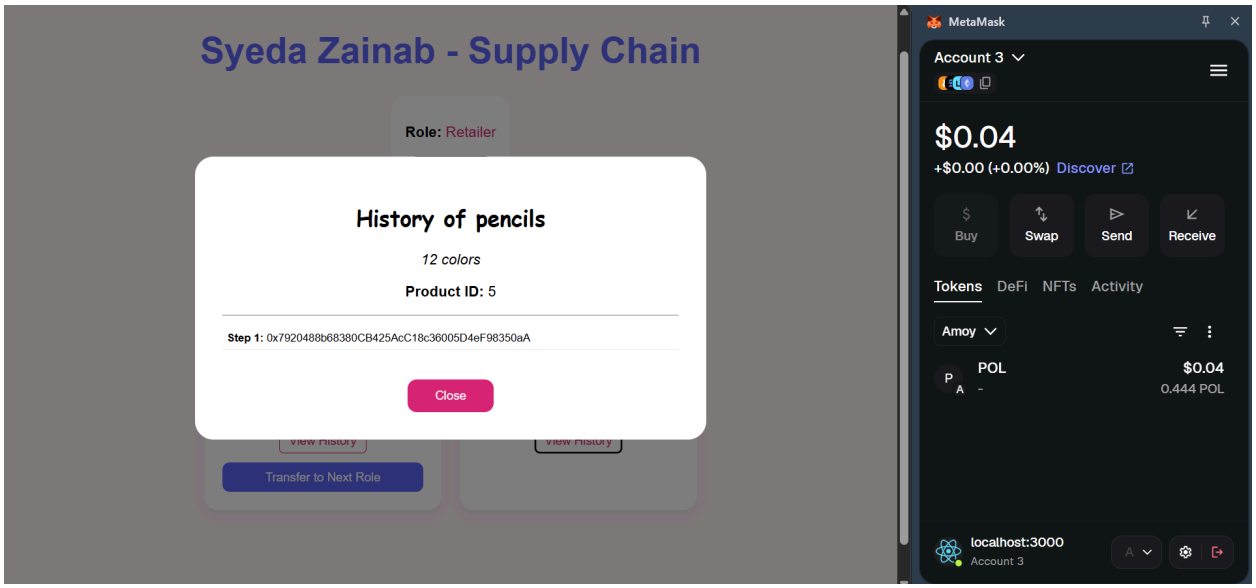


G. Retailer Receipt & Audit Trail

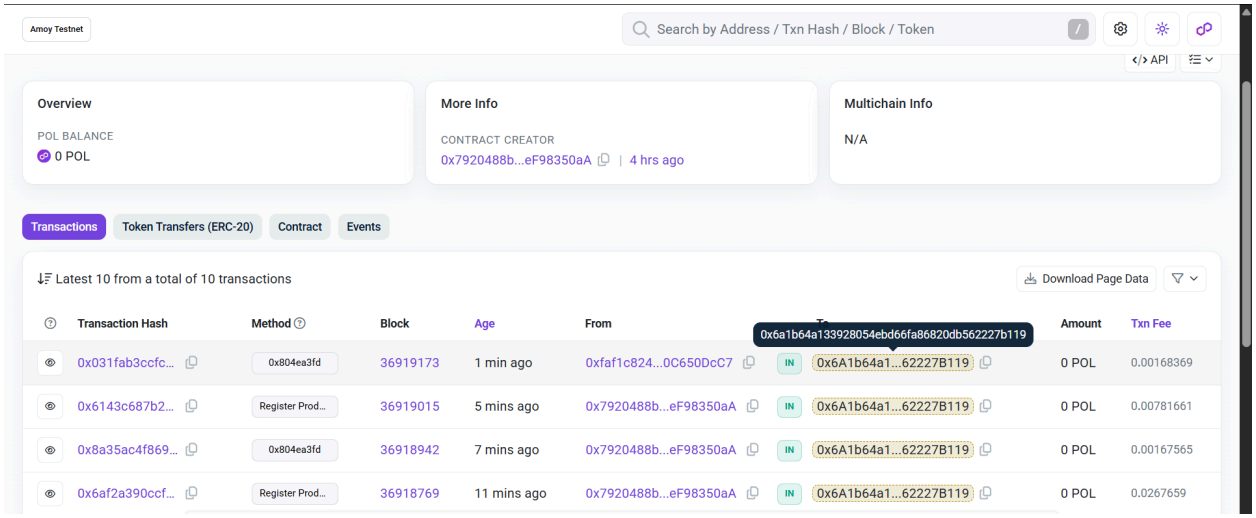
- The snapshot shows the product has reached the **Retailer**(Account 3).



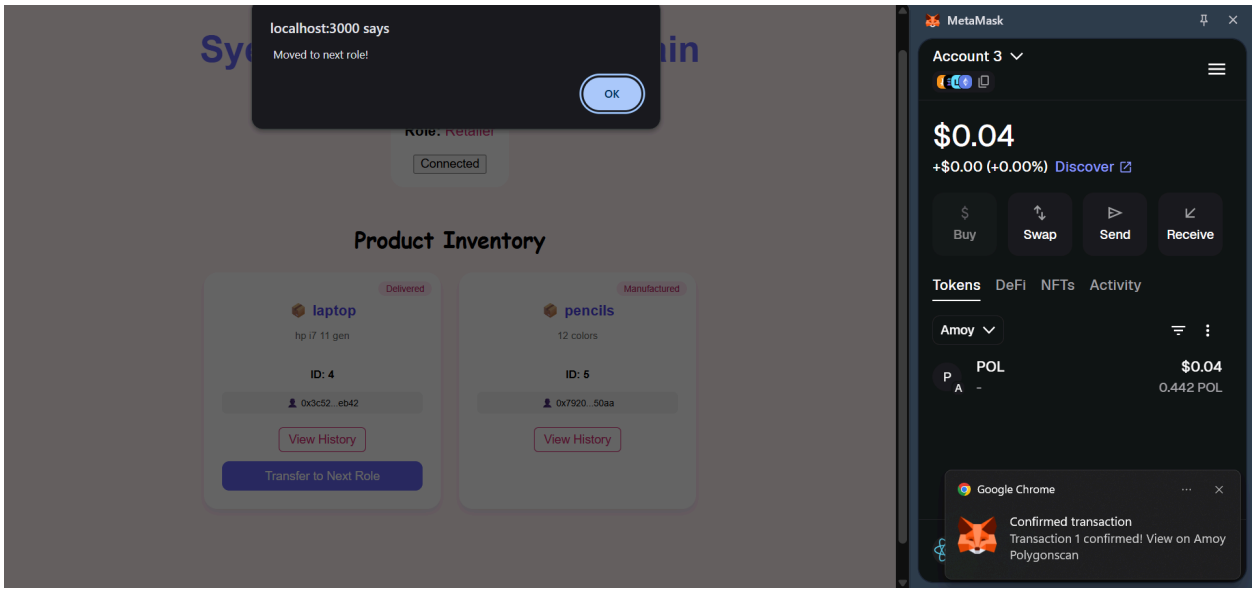
- The "History of laptop" model displays a complete **3-step audit trail**, showing every wallet address that handled the product from manufacture to retailer.



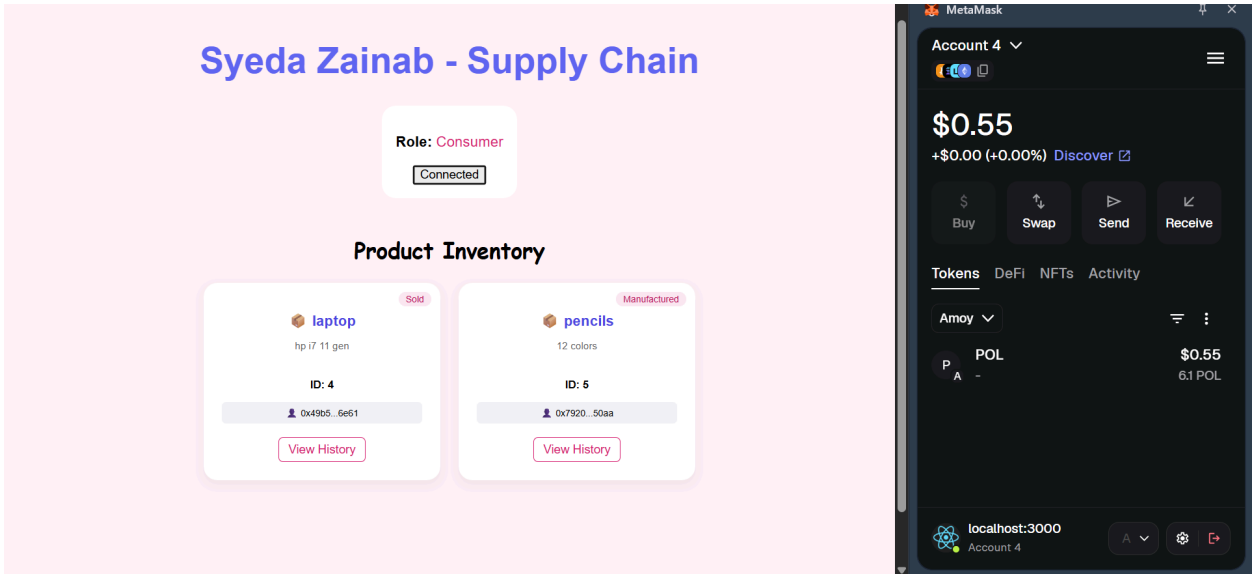
- As product with ID=5 (pencils) has not been transferred , it shows the address of manufacturer only.

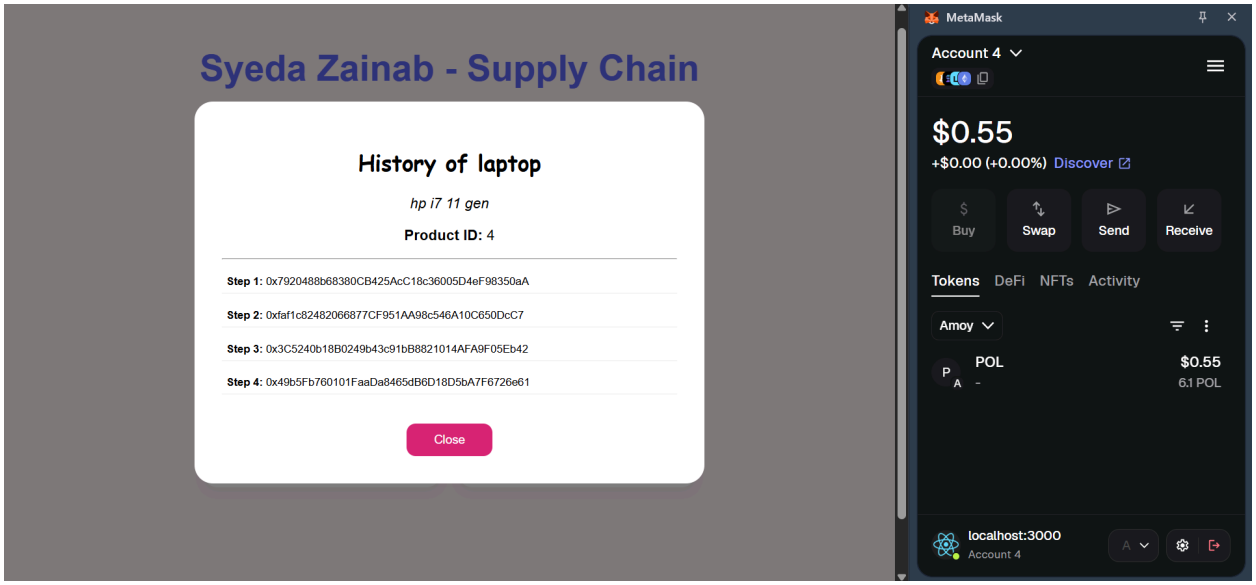


- Now account 3(retailer) transfers ownership of the laptop to consumer (account 4).



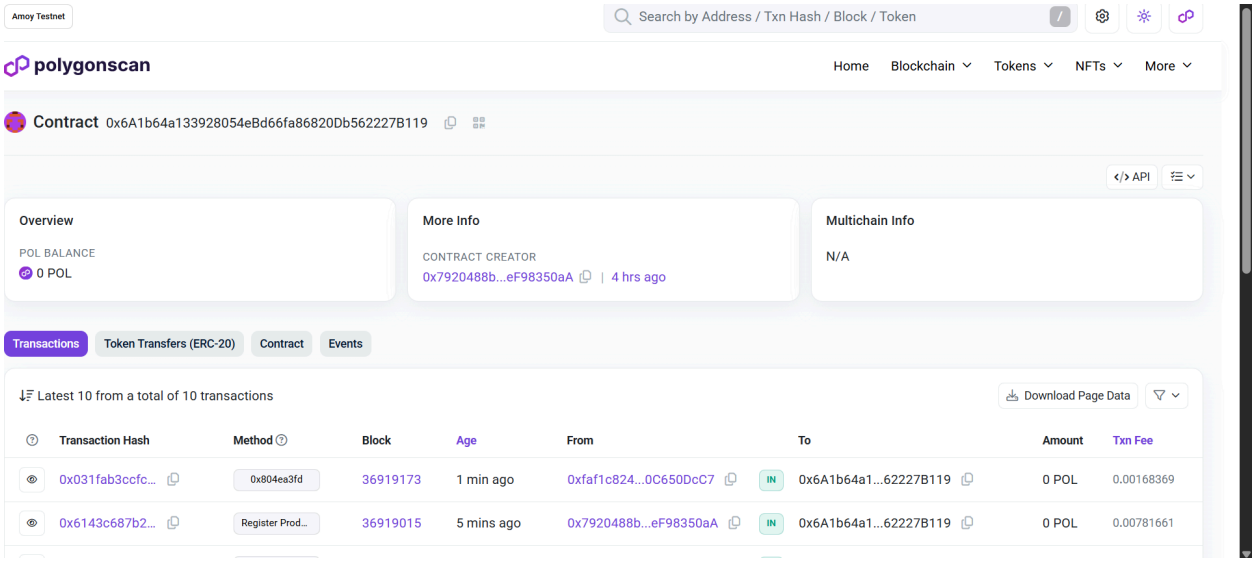
- The product (laptop) appears to be sold here, as it has completed the cycle from distributor to consumer .





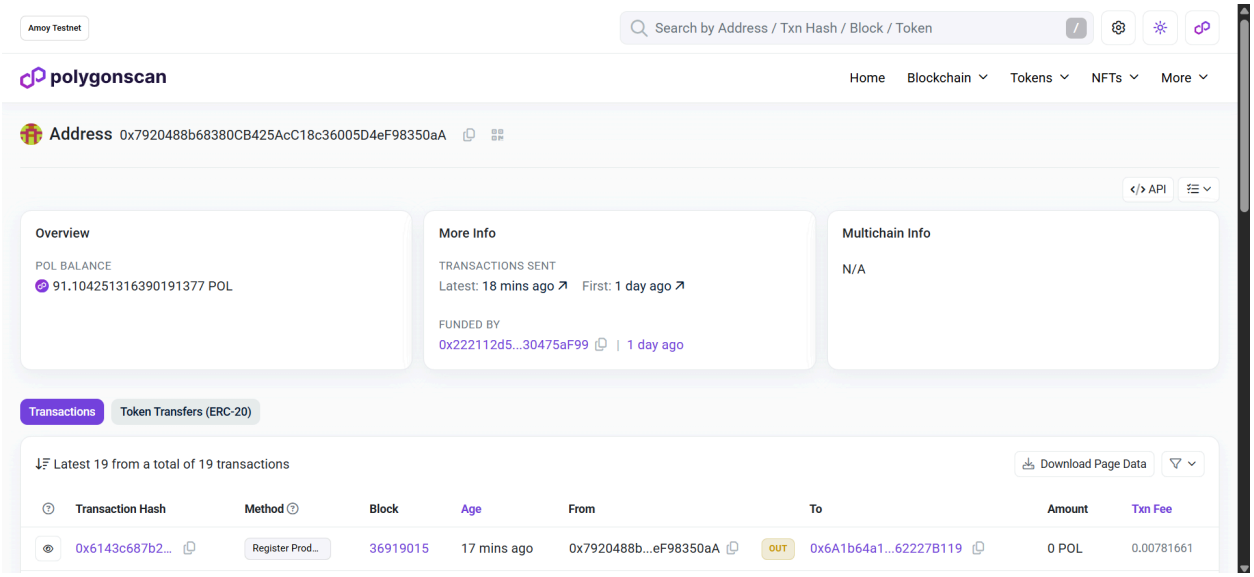
G. Final Consumer Receipt & Full Audit Trail

- The final snapshot shows the product has reached the **Consumer** (Account 4). The "History of laptop" modal displays a complete **4-step audit trail**, showing every wallet address that handled the product from manufacture to final sale.



Conclusion

The system successfully demonstrates the core principles of blockchain in supply chain management:

1. Transparency: All transactions are visible on Polygonscan.
2. Role Security: Only authorized owners can move products.
3. Traceability: A full history is maintained for every unique Product ID.

This is the detail of my account 2 manufacturer